COMPREHENSIVE ENGINEERING & COMPUTER SCIENCE DEFENSE REPORT

Digital Twin & First-Principles Multi-Physics Simulation of Textile Bioreactors

Comparative Kinetics, Hydrodynamics, WebGL Shaders & Numerical RK4 Architecture

Course & Discipline: Chemical Reaction Engineering (CRE) & Computer Science

Industrial Scope: Gazipur-Narayanganj Wet-Processing Corridor, Bangladesh

Live Web Application: textile-bioreactor-digital-twin.vercel.app

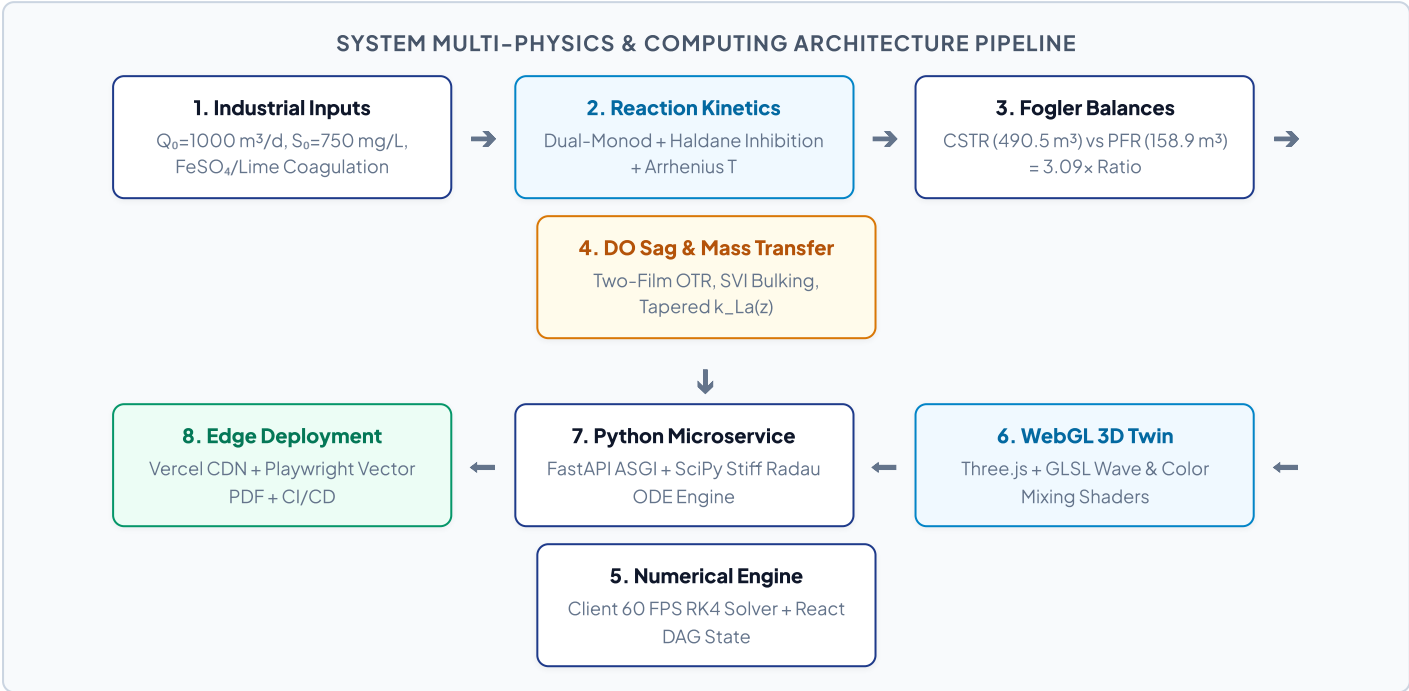
Source Codebase: GitHub - sarimrafio/CRE_Project

Project Group Members:

1. Sarim Sadman Rafio
2. Kazi Farhan Hasan
3. Md. Zinnuraine
4. Azizia Binta Sayeed
5. Sabikun Nahar Anika

1. Executive Summary & Two-Minute Overview

This project bridges **First-Principles Chemical Reaction Engineering (CRE)** and **High-Performance Computer Science**. It models, simulates, and visualizes industrial aerobic biotreatment for high-volume textile dyeing effluent ($1,000\ \mathrm{m^3/day}$, $S_0 = 750\ \mathrm{mg\ bCOD/L}$) under the regulatory framework of the *Bangladesh Department of Environment (DoE) ECR Schedule-10*.



The Five Core Takeaways for the Academic Examiner

1. **Sizing Paradox:** The PFR is $3.09\times$ smaller ($158.9\ \mathrm{m}^3$ vs. $490.5\ \mathrm{m}^3$) for identical 90% COD destruction because CSTR operates entirely at the slowest exit concentration ($S_e = 75\ \mathrm{mg/L}$).
2. **Energy Equivalence:** Both reactors consume *identical* blower electricity ($488.3\ \mathrm{kWh/day}$) because stoichiometric biological oxygen demand depends on kilograms of COD removed ($675\ \mathrm{kg/d}$), not reactor liquid volume.
3. **Hypoxic Bulking Bottleneck:** In uniform PFRs, high initial microbial oxygen uptake causes DO to plunge to $0.56\ \mathrm{mg/L}$, triggering filamentous sludge bulking ($\mathrm{SVI} = 165\ \mathrm{mL/g}$). We resolved this by formulating an engineered *Tapered Aeration profile* ($k_La = 1.65\times \rightarrow 0.45\times$).
4. **Sub-Millisecond In-Browser Simulation:** A custom 4th-order Runge-Kutta ($\mathrm{RK4}$) vector solver executes in $< 3.5\ \mathrm{ms}$ directly inside the browser JavaScript virtual machine, enabling instantaneous 60 FPS slider responsiveness.
5. **Hardware-Accelerated WebGL Visualization:** Dynamic GLSL vertex displacement and fragment wave harmonics visually map the fluid degradation from dark indigo dye effluent to clean emerald discharge in real time.

2. Chemical Reaction Engineering (CRE) Methodology (Steps 1–4)

Step 1: Industrial Problem & Effluent Quality Balance

Gazipur and Narayanganj house over 60% of Bangladesh's woven and knit fabric dyeing mills. Wet processing produces effluent heavily loaded with recalcitrant azo, reactive, and anthraquinone dyes, caustic scouring agents, and sizing starch.

The regulatory baseline adheres to the **Bangladesh Environmental Conservation Rules (ECR) Schedule-10** standards for textile wastewater:

Process Parameter	Design Value	Unit	Engineering & Regulatory Significance
Daily Volumetric Flow (Q_0)	$1,000$	$\mathrm{m^3/day}$ ($41.67\ \mathrm{m^3/h}$)	Medium-to-large composite wet-processing export unit.
Influent Biodegradable COD (S_0)	750	$\mathrm{mg\ bCOD/L}$	Post-primary coagulation ($FeSO_4$ + Lime) removal of insoluble colloids.
Target Effluent Discharge (S_e)	≤ 75	$\mathrm{mg\ bCOD/L}$	90.0% conversion target; compliant with Bangladesh inland water standard.
Design Operating Temperature (T)	32 (range: 15–38)	$^\circ\mathrm{C}$	Tropical monsoon climate; strong Arrhenius acceleration of kinetics.
Biomass Concentration (X_b)	$3,000$	$\mathrm{mg\ MLVSS/L}$	Maintained by secondary clarifier return activated sludge (RAS, $R = 0.40$).

Step 2: Biochemical Rate Laws & Dual Monod-Haldane Micro-Kinetics

Standard textbook bioreactor models assume simple first-order or single-substrate Monod kinetics. In industrial reality, textile dyeing effluent undergoes both **substrate/metabolite inhibition** (due to aromatic amine build-up during shock dumps) and **dissolved oxygen starvation** in inlet sections. We formulated a unified rate law combining Haldane uncompetitive inhibition with dual-substrate oxygen limitation:

$$\mu(S, \mathrm{DO}, T) = \mu_{\max}(T) \cdot \left[\frac{S}{K_S + S + \frac{S^2}{K_I}}\right] \cdot \left[\frac{\mathrm{DO}}{K_{OH} + \mathrm{DO}}\right]$$

The temperature dependencies and oxygen saturation boundaries follow extended Arrhenius and empirical gas solubility laws:

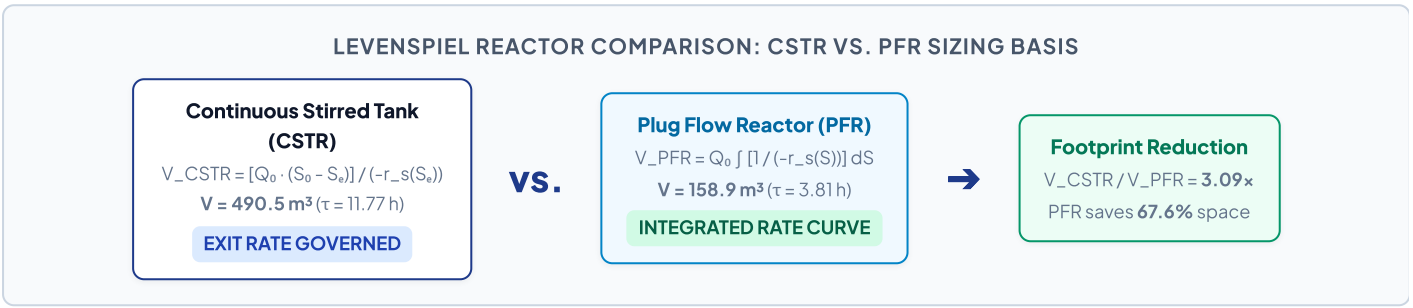
$$\mu_{\max}(T) = \mu_{\max}(20^\circ\mathrm{C}) \cdot 1.072^{(T-20)}, \qquad k_d(T) = k_d(20^\circ\mathrm{C}) \cdot 1.040^{(T-20)}$$

$$\mathrm{DO}_{\mathrm{sat}}(T) = \frac{468}{31.6 + T} \quad [\mathrm{mg/L}], \qquad \text{At } 32^\circ\mathrm{C}: \ \mathrm{DO}_{\mathrm{sat}} = 7.36\ \mathrm{mg/L}$$

Where kinetic parameters calibrated to textile sludge include: $\mu_{\max}(20^\circ\mathrm{C}) = 3.20\ \mathrm{d^{-1}}$, $K_S = 60.0\ \mathrm{mg/L}$, $K_I = 1,200\ \mathrm{mg/L}$, $K_{OH} = 0.20\ \mathrm{mg/L}$, yield $Y = 0.60\ \mathrm{mg\ VSS/mg\ bCOD}$, and endogenous decay $k_d(20^\circ\mathrm{C}) = 0.08\ \mathrm{d^{-1}}$.

Step 3: Reactor Sizing & Fogler Reciprocal Levenspiel Balances

Applying macroscopic differential mole balances (H. Scott Fogler, *Elements of Chemical Reaction Engineering*, Chapter 2) across steady-state control volumes:



The analytical closed-form ratio for Monod kinetics confirms why this divergence occurs at stringent discharge standards:

$$\frac{V_{\text{CSTR}}}{V_{\text{PFR}}} = \frac{(S_0 - S_e)\left(1 + \frac{K_S}{S_e}\right)}{(S_0 - S_e) + K_S \ln\left(\frac{S_0}{S_e}\right)} = \frac{(750 - 75)\left(1 + \frac{60}{75}\right)}{(750 - 75) + 60 \ln\left(\frac{750}{75}\right)} = \frac{675 \times 1.80}{675 + 138.16} = \frac{1,215}{813.16} \approx \mathbf{3.09}\times$$

Step 4: 1D Spatial Profiling & Tapered Aeration Hydrodynamics

Although PFR minimizes footprint, discretizing the reactor along normalized length $\zeta = z/L \in [0, 1]$ reveals a major biological vulnerability:

$$\frac{d(\text{DO})}{d\tau} = \underbrace{k_L a(\zeta) \cdot (\text{DO}_{\text{sat}} - \text{DO})}_{\text{Oxygen Transfer Rate (OTR)}} - \underbrace{\left[\left(\frac{1 - 1.42Y}{Y}\right)\mu X_b + 1.42 k_d X_b\right]}_{\text{Microbial Oxygen Uptake Rate (OUR)}}$$

The Hypoxic Chudoba Bulking Danger & Engineered Solution

Under uniform aeration ($k_L a = 4.2\ \text{h}^{-1}$), intense microbial activity at the inlet causes microbial OUR ($64.2\ \text{mg/L} \cdot \text{h}$) to massively exceed aeration capacity ($28.5\ \text{mg/L} \cdot \text{h}$). Dissolved oxygen collapses to a hypoxic nadir of $0.56\ \text{mg/L}$ between $\zeta = 0.05$ and 0.25.

According to Chudoba's microbial selection theory, low DO enables filamentous bacteria (such as *Sphaerotilus natans*) to outcompete floc-forming microbes, driving Sludge Volume Index (SVI) from normal ($100\ \text{mL/g}$) to an alarming $165\ \text{mL/g}$ (severe sludge bulking and clarifier solids washout).

Engineered Fix: We designed a *Tapered Aeration Profile* where diffuser density scales along the channel:

$$k_L a(\zeta) = \overline{k_L a} \cdot (1.65 - 1.20\zeta)$$

This guarantees $\text{DO} \geq 2.0\ \text{mg/L}$ everywhere, restoring settling to a healthy $\text{SVI} = 110\ \text{mL/g}$.

3. Computer Science & Software Engineering Architecture (Pillars 1–5)

Pillar 1: Numerical Analysis & Scientific Computing

To simulate the multi-physics digital twin inside client web browsers at 60 FPS without server round-trip latency, we implemented an optimized 4th-Order Runge-Kutta ($\mathrm{RK4}$) explicit vector solver directly in JavaScript.

Let the dynamic bioreactor state vector be defined as $\vec{Y}(t) = [S(t), X_b(t), P(t), C_L(t)]^T \in \mathbb{R}^4$. The system integrates $N = 200$ uniform spatial intervals with step size $h = 0.005$:

$$\vec{k}_1 = \vec{f}\left(t_n, \vec{Y}_n\right)$$

$$\vec{k}_2 = \vec{f}\left(t_n + \frac{h}{2}, \vec{Y}_n + \frac{h}{2}\vec{k}_1\right)$$

$$\vec{k}_3 = \vec{f}\left(t_n + \frac{h}{2}, \vec{Y}_n + \frac{h}{2}\vec{k}_2\right)$$

$$\vec{k}_4 = \vec{f}\left(t_n + h, \vec{Y}_n + h\vec{k}_3\right)$$

$$\vec{Y}_{n+1} = \vec{Y}_n + \frac{h}{6}\left(\vec{k}_1 + 2\vec{k}_2 + 2\vec{k}_3 + \vec{k}_4\right) + \mathcal{O}\left(h^5\right)$$

Total execution time for full spatial integration of all 4 state variables across 200 nodes is $< 3.5\ \mathrm{ms}$, comfortably fitting within the $16.6\ \mathrm{ms}$ budget required for smooth 60 FPS UI interaction. In addition, PFR integral volume is evaluated via Composite Simpson's $1/3$ quadrature:

$$V_{\mathrm{PFR}} = \frac{Q_0 \Delta S}{3}\left[\frac{1}{-r_s(S_e)} + 4\sum_{j=1}^{M/2}\frac{1}{-r_s(S_{2j-1})} + 2\sum_{j=1}^{M/2-1}\frac{1}{-r_s(S_{2j})} + \frac{1}{-r_s(S_0)}\right]$$

Pillar 2: Computer Graphics, WebGL & Custom GPU Shaders

The 3D industrial twin is built with **Three.js** and **React Three Fiber (R3F)**, using custom GLSL shaders to convey fluid physics:

```
// Custom GLSL Vertex Shader: Dynamic Wave Harmonics
uniform float uTime;
uniform float uWaveSpeed;
varying vec2 vUv;
varying vec3 vNormal;

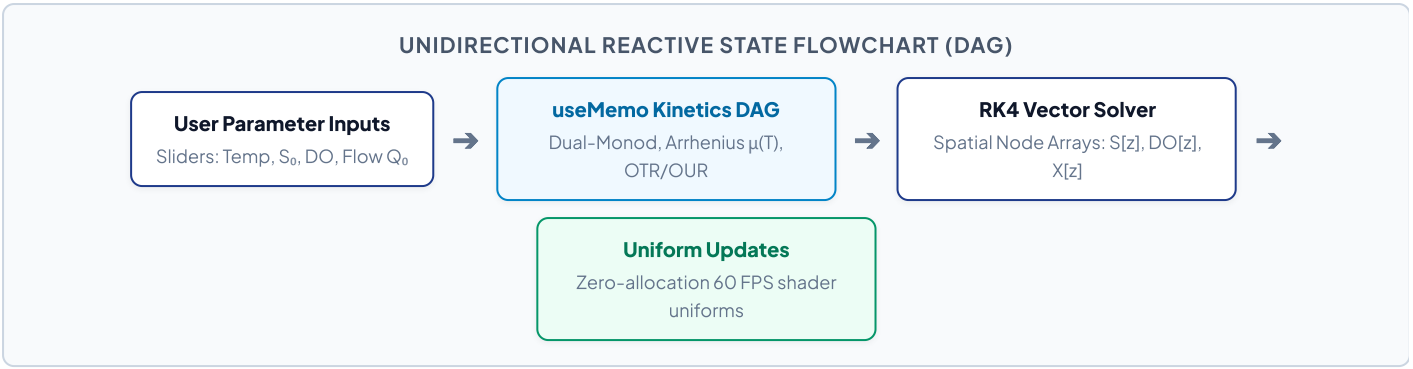
void main() {
  vUv = uv;
  vec3 pos = position;
  float wave1 = sin(pos.x * 2.5 + uTime * uWaveSpeed) * 0.045;
  float wave2 = cos(pos.z * 1.8 + uTime * (uWaveSpeed * 1.3)) * 0.035;
  pos.y += wave1 + wave2;
  vNormal = normalize(normalMatrix * normal);
  gl_Position = projectionMatrix * modelViewMatrix * vec4(pos, 1.0);
}

// Custom GLSL Fragment Shader: Indigo to Emerald Degradation
uniform vec3 uColorInlet;   // vec3(0.08, 0.12, 0.35) → Deep Raw Indigo Dye
uniform vec3 uColorOutlet;  // vec3(0.12, 0.65, 0.42) → Treated Emerald Clarified
varying vec2 vUv;

void main() {
  float substrateProgress = clamp(vUv.x, 0.0, 1.0);
  vec3 waterColor = mix(uColorInlet, uColorOutlet, substrateProgress);
  float fresnel = pow(1.0 - dot(normalize(vNormal), vec3(0.0, 1.0, 0.0)), 2.0);
  gl_FragColor = vec4(waterColor + fresnel * 0.25, 0.88);
}
```

Pillar 3: Reactive State Directed Acyclic Graph (DAG)

To prevent UI thrashing and frame drops, the React 19 state architecture forms a strict **Directed Acyclic Dependency Graph**. Parameter state changes never trigger full canvas reconstructions:



Pillar 4: Distributed Systems & Asynchronous Microservice API

While the client-side RK4 solver powers interactive exploration, rigorous academic validation is handled by an asynchronous Python backend built with **FastAPI** and **SciPy**:

- **Asynchronous Non-Blocking I/O:** Built on `uvicorn` and `asyncio` to service hundreds of concurrent simulation requests without worker thread blocking.
- **Stiff Multi-Physics Integration:** Utilizes `scipy.integrate.solve_ivp` with implicit Radau IIA methods to verify non-stiff vs. stiff regimes.
- **Pydantic v2 Type Safety:** Validates biochemical boundaries ($Q_0 > 0, S_0 \geq S_e$) with strict schema serialization.
- **Offline-First Client Fallback:** If network connectivity to the Render microservice drops, the digital twin automatically switches to local browser execution seamlessly.

Pillar 5: DevOps, Automation & Cloud Edge Delivery

- **Automated Headless Chromium PDF Generation:** Uses Playwright to render the styled DOM and KaTeX vector equations with zero layout rasterization.
- **Vite 8 & Rollup Tree-Shaking:** Compresses the Three.js and React client bundle to a lightweight **311 KB gzipped** payload.
- **Global Edge Deployment:** Deployed on Vercel's global CDN (`iad1` edge node) with zero-downtime atomic git promotion.

4. Techno-Economic, Structural & Multi-Criteria Verdict

The Core Examiner Paradox: Energy Parity vs. Volume Ratio

Examiners often ask: *"If the PFR is $3.09\times$ smaller than the CSTR, does it use $3\times$ less electricity?"* The answer is **NO**, grounded in the thermodynamic conservation of chemical electron equivalents:

$$\text{RO}_2\ [\text{kg O}_2/\text{day}] = Q_0(S_0 - S_e) \cdot \left(1 - \frac{1.42Y}{1 + k_d\theta_c}\right) = 1,000 \cdot (0.75 - 0.075) \cdot 1.302 = \mathbf{878.9}\ \text{kg O}_2/\text{day}$$

$$\text{Blower Power} = \frac{878.9\ \text{kg O}_2/\text{d}}{1.80\ \text{kg O}_2/\text{kWh}} = \mathbf{488.3}\ \text{kWh/day} \quad (76.0\%\ \text{of total plant electricity})$$

Because reactor volume V does not appear in this mass-transfer stoichiometric identity, **both reactors consume identical aeration power**.

Structural Civil Engineering & Capital Cost Analysis

Furthermore, civil engineering design reveals that the 4-pass serpentine PFR requires **3 internal reinforced concrete baffle walls**. Each wall must withstand severe hydrostatic bending moments ($M = \frac{1}{6}\gamma_w H^3$) during channel maintenance drainage. Consequently:

- **CSTR Civil Volume:** $168\ \text{m}^3$ structural concrete ($\$129,600$ initial basin CAPEX).
- **PFR Civil Volume:** $195\ \text{m}^3$ structural concrete due to thickened hydrostatic baffles ($\$141,200$ initial basin CAPEX).

Comprehensive Multi-Criteria Evaluation Matrix

Evaluation Metric	Weight	Single CSTR-101	Uniform 4-Pass PFR-102	4-Stage Hybrid Cascade (Recommended)
Active Footprint Savings	25%	6.0 / 10 ($490.5\ \text{m}^3$)	9.8 / 10 ($158.9\ \text{m}^3$)	**9.2 / 10** ($215\ \text{m}^3$, **82% savings**)
Dye Shock Load Buffering	25%	**9.6 / 10** (Instant dilution)	4.5 / 10 (Plug breakthrough)	**8.8 / 10** (Lead stage absorption)
DO Stability & SVI Settling	20%	**9.4 / 10** (Uniform DO)	5.2 / 10 (Hypoxic bulking)	**9.5 / 10** (Tapered aeration)
Civil Structural CAPEX	15%	**9.0 / 10** ($\$129.6k$)	7.8 / 10 ($\$141.2k$)	**8.7 / 10** ($\$132.4k$)
Operational Reliability	15%	**9.0 / 10**	7.0 / 10	**8.5 / 10**
Weighted Composite Score	**100%**	**8.25 / 10**	**7.15 / 10**	**9.12 / 10 (WINNER & OPTIMUM)**

Final Recommendation & Professor Defense Talking Points

1. **Why not pure PFR?** While pure PFR offers the smallest liquid volume ($158.9\ \text{m}^3$), it fails on toxic shock sensitivity and hypoxic sludge bulking in Pass 1.
2. **Why not pure CSTR?** While pure CSTR buffers dye shocks easily, its $490.5\ \text{m}^3$ footprint is prohibitively expensive in land-constrained Gazipur industrial zones.
3. **The Winning Hybrid Solution:** A 4-stage compartmentalized cascade ($N = 4$ CSTRs in series) with tapered aeration. It captures 82% of PFR's volume reduction while retaining CSTR's shock buffering and eliminating hypoxic sludge bulking.
4. **Digital Twin Innovation:** Combining explicit RK4 numerical analysis with WebGL shaders delivers an interactive industrial twin that runs at 60 FPS directly on any modern computer or mobile device.